

A bounded pulse comparison: the released Navier–Stokes wave ODE and the coupled viscous pair

1 Scope and source equations

This note records one local, exact comparison. It does not identify the coupled Boussinesq stages with the released Navier–Stokes construction and makes no claim about a global Navier–Stokes solution. The source is the released PDF `openai_navier_stokes.pdf`, SHA-256 `8c8a94ad9ac824c8b605b9827cacf7beaca48bd10b380de3cfc872a2c37afa81`. The source statements used here are Section 7, equations (7.2)–(7.8), (7.12)–(7.22), and Section 9, equations (9.1)–(9.2), on PDF pages 74–75, 78–81, and 102–103. In the source notation, on one lifted pulse rectangle,

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v(pF + p_z G), \quad n = \nabla_* \Phi, \quad (1)$$

with

$$n = (x_0 - v(pF_R + p_z G_R), \ p/R, \ p_z - \varepsilon v(pF_Z + p_z G_Z)). \quad (2)$$

For a nonzero integer harmonic m , the exact principal pulse equations are

$$n \cdot t_m = 0, \quad t'_m + K t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad d_N = \varepsilon k^2 |n|^2, \quad (3)$$

where a prime is the pulse-coordinate derivative and

$$K = \begin{pmatrix} 0 & -2F & 0 \\ 2F + R F_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad A_\Phi = -K + \frac{n(n^T K - (n')^T)}{|n|^2}. \quad (4)$$

The source pressure coefficient is exactly

$$\pi_m = \frac{i}{k m |n|^2} (n \cdot K t_m - (n') \cdot t_m + n \cdot f_m). \quad (5)$$

The source's energy identity (7.22) is

$$\frac{1}{2} (|t_m|^2)' = -g \cdot (t_{m,r}(t_{m,\theta}, t_{m,z})) - \varepsilon k^2 |n|^2 |t_m|^2, \quad (6)$$

where $g = (R F_R, G_R)$ in the tangential coordinates. Thus the source pulse retains both shear transfer and viscous dissipation.

2 The coupled physical pair

For one activated layer of the coupled calculation, put

$$a_c = -\frac{J\zeta \cdot G_{<q}}{Kr^2}, \quad b_c = K\zeta_1, qquad \lambda_c^\Theta = \kappa K^2 r^2, qquad \lambda_c^\Omega = \varepsilon_c K^2 r^2. \quad (7)$$

Its exact retained-viscosity pair, including optional scalar and vorticity controls, is

$$\dot{y} = C_c y + c_c, qquad y = \begin{pmatrix} \Theta \\ \Omega \end{pmatrix}, \quad C_c = \begin{pmatrix} -\lambda_c^\Theta & a_c \\ b_c & -\lambda_c^\Omega \end{pmatrix}, \quad c_c = \begin{pmatrix} c_\theta \\ c_\omega \end{pmatrix}. \quad (8)$$

Here $G_{<q}$, ζ , and r evolve with the inherited background and phase ODEs; in particular no fixed-background substitution is made. Let v be a source pulse coordinate on an interval and let $q_v = dt/dv > 0$ be its physical-time parametrization. Define $y'(v) = q_v(C_c y + c_c)$.

3 Exact local residual map

Let $B(v) \in \mathbb{R}^{3 \times 2}$ be a C^1 tangent frame with full column rank and $n(v)^T B(v) = 0$, so its range is exactly $n(v)^\perp$. Let $B_\ell(v)$ be any left inverse, $B_\ell B = I_2$; then BB_ℓ acts as the identity on $n^\perp$. Let $L(v) \in \mathbb{R}^{2 \times 2}$ be any C^1 coefficient map and embed the coupled pair into a transverse source amplitude by

$$t(v) = B(v)L(v)y(v). \quad (9)$$

For an embedding with a non-tangent frame the transversality condition would be

$$n(v)^T B(v)L(v)y(v) = 0. \quad (10)$$

For the tangent frame used here, the condition holds for every L, y . Set

$$H(v) = B_\ell(v)(A_\Phi(v)B(v) - B'(v)). \quad (11)$$

Then the exact projected source residual of the embedded coupled pair is

$$\mathcal{E}_{L,c} := B_\ell \text{proj}_{n^\perp} [t' - A_\Phi t + m^2 d_N t + f_m] = [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \quad (12)$$

$$+ B_\ell \text{proj}_{n^\perp} f_m. \quad (13)$$

In particular, with $L = I$ and no added source, the defect is the explicit matrix expression

$$\mathcal{E}_{I,0} = (q_v C_c - H + m^2 d_N I_2) y. \quad (14)$$

This is the strongest local relation available without asserting that the two constructions have identical frames, coordinates, or coefficients.

Proof. Differentiate (9): $t' = B' L y + B L' y + B L y'$. Since $y' = q_v(C_c y + c_c)$, left-multiply $t' - A_\Phi t + m^2 d_N t + f_m$ by B_ℓ and use $B_\ell B = I_2$. The terms involving B' and $A_\Phi B$ combine as $-H L$ by (11); the remaining terms are exactly those in (13). Because A_Φ is the projected operator,

$$n^T(t' - A_\Phi t + m^2 d_N t + f_m) = 0 \quad (15)$$

whenever $n^T B = 0$ and the forcing pressure below is used. Hence the projected identity determines the full residual in the two-dimensional transverse plane. $\square$

4 Pressure reconstruction and full three-dimensional residual

For any embedded transverse t and arbitrary f_m , define

$$\pi_{L,c} = \frac{i}{km|n|^2} (n \cdot \mathbf{K}t - (n') \cdot t + n \cdot f_m). \quad (16)$$

Then the full source residual satisfies the exact decomposition

$$\mathcal{R}_{L,c} := t' + \mathbf{K}t + m^2 d_N t + ikmn\pi_{L,c} + f_m \quad (17)$$

$$= \text{proj}_{n^\perp} (t' + \mathbf{K}t + m^2 d_N t + f_m) = B \mathcal{E}_{L,c}, \quad (18)$$

where the last equality uses the definition of H and the transverse condition. Thus the pressure removes precisely the normal component; it does not remove $\mathcal{E}_{L,c}$, the moving-frame, shear, diffusion, or coupled-background mismatch. If one elects to force the source pulse to obey the released equation, then the unique transverse forcing represented in the frame is

$$f_{m,\text{corr}} = -B \mathcal{E}_{L,c}, \quad (19)$$

with pressure still given by (16); this is an exact local forced relation, not an identification of the physical constructions.

5 Diffusion and energy comparison

The source diffusion multiplier is $m^2 d_N = m^2 \varepsilon k^2 |n|^2$ in (3). The coupled pair has the two retained physical losses λ_c^Θ and λ_c^Ω , which are generally unequal to $m^2 d_N$. Their signed discrepancy in (14) is explicitly

$$(m^2 d_N - q_v \lambda_c^\Theta) \Theta, \quad (m^2 d_N - q_v \lambda_c^\Omega) \Omega \quad (20)$$

when $H = 0$ and C_c is expanded entrywise; with general H the exact diagonal entries are those of $q_v C_c - H + m^2 d_N I_2$. No damping is dropped. If one compares nonnegative loss magnitudes instead, the corresponding cost gaps are $|m^2 d_N - q_v \lambda_c^\Theta|$ and $|m^2 d_N - q_v \lambda_c^\Omega|$. If a coordinate map L is chosen to solve

$$L' = HL - q_v L C_c - m^2 d_N L, \quad (21)$$

then the homogeneous embedded pair has zero projected residual, but this is a new local coefficient transformation and does not make the source phase, pressure, cutoff, or nonlinear covariance equal to the coupled Boussinesq fields. For general controls c_c , the forced term is $q_v L c_c$ in (13).

For the embedded field, the source energy balance is exactly

$$\frac{1}{2} (|t|^2)' = -t \cdot \mathbf{K}t - m^2 d_N |t|^2 \quad (22)$$

when $f_m = 0$ and the transversality constraint is imposed. Since $t \cdot \mathbf{K}t = g \cdot (t_r(t_\theta, t_z))$, this is precisely (6). The coupled pair has instead

$$\frac{1}{2} \frac{d}{dt} |y|^2 = -\lambda_c^\Theta \Theta^2 - \lambda_c^\Omega \Omega^2 + (a_c + b_c) \Theta \Omega + c_\theta \Theta + c_\omega \Omega, \quad (23)$$

which is a two-component physical energy identity. Equations (13) and (23) exhibit exactly where the source's three-dimensional shear flux and the coupled pair's two-dimensional exchange differ.

6 Proven scope

The source uses a cylindrical, localized, divergence-free curl construction (Section 7.7 and Section 9, PDF pages 88–89 and 102–103); the displayed principal pulse equation is only one component of that construction. The comparison above proves an exact local map and its full pressure-projected residual for every finite interval on which the denominators and frame left inverse exist. It preserves source viscosity, moving phase normal, pressure, rotation/shear matrix, and all coupled physical diffusion coefficients. It does not prove that the coupled third-growth or steering stages produce the source’s pulse phase Φ , covariance, cutoff tails, or smooth terminal force. Those remain separate bridge problems.

The sign conventions in this comparison are also explicit. The quantities $q_v > 0$, $\varepsilon > 0$, and $d_N = \varepsilon k^2 |n|^2 > 0$ record the chosen orientation of physical time and the positive viscous coefficient; they are not assumptions on the sign of a pulse or blowout amplitude. The equations (3)–(5) are linear in (t_m, f_m) , so the simultaneous replacement $(t_m, f_m, \pi_m) \mapsto (-t_m, -f_m, -\pi_m)$ is an exact reflected branch. Likewise, the local map (9) is sign-equivariant under $y \mapsto -y$ and $c_c \mapsto -c_c$. The source’s positive covariance weights are quadratic bookkeeping for its selected construction and do not rule out a negative velocity branch. No conclusion about the sign of a blowout is therefore drawn from the positive geometric or viscosity scales.